Difference between boolean and binary... Mean 1+=1 while in binary 1+1=10

The difference is Boolean logic vs Binary arithmetic.

1. Boolean (Logic)

Used in conditions, switches, true/false decisions.

Values are only:

- 0 = False
- 1 = True

Here + usually means OR (not normal addition).

So:

- 1 + 1 = 1 → because True OR True = True
- 1 · 1 = 1 → AND
- 1' = 0 → NOT

Example: In programming `if`, `while`, digital gates.

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2. Binary (Number System)

Used for numbers and calculations in base-2.

Digits:

- 0
- 1

Here + means real arithmetic addition.

So:

- 1 + 1 = 10 (binary)
  Because decimal 1 + 1 = 2, and 2 in binary = 10

- 10 + 1 = 11
- 11 + 1 = 100

Easy Memory Trick

- Boolean = Thinking / Decision / Logic
- Binary = Counting / Numbers / Math

So yes, exactly:

- Boolean: 1 + 1 = 1
- Binary: 1 + 1 = 10

Explain in more and simple easy details along with example

Yes — this confuses many people because both use only 0 and 1, but they mean different things.

Main Difference

Type	Used For	Meaning of 0 and 1
Boolean	Logic / decisions	0 = False, 1 = True
Binary	Numbers / counting	0 and 1 are digits

So same symbols 0 and 1, but different purpose.

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1. Boolean (Logic System)

Boolean is based on true/false decisions.

- 0 = False
- 1 = True

Used in:

- Conditions in programming
- Digital logic gates

- Switch ON/OFF systems

Example 1: Light Switch

If switch is ON = 1

If switch is OFF = 0

Example 2: Login Check

If password correct = 1

If wrong = 0

Boolean Addition (+)

In Boolean algebra, + does NOT mean normal math addition.

It means OR.

So:

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

So:

1 + 1 = 1

Because:

True OR True = True

Real Example

If you can enter room when:

- You have key = 1
  OR
- You have password = 1

Then:

1 + 1 = 1
(You still can enter, not double enter)

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2. Binary (Number System)

Binary is used for numbers.

Only digits:

- 0
- 1

Just like decimal uses:
0,1,2,3,4,5,6,7,8,9

Binary uses base 2.

Binary Counting

Decimal	Binary
0	0
1	1
2	10
3	11
4	100

Binary Addition

Here + means real math.

Example:

1 + 1 = 2 decimal

But in binary, decimal 2 = 10

So:

1 + 1 = 10

More Binary Examples

1 + 0 = 1

10 + 1 = 11

11 + 1 = 100

Why Same Symbols But Different Answer?

Because:

- Boolean is logic rules
- Binary is number rules

Like word "bat" can mean animal or cricket bat depending on context.

Same 1 and 0, different meaning.

Computer Uses Both

A CPU uses:

- Binary for calculations (numbers)
- Boolean for decisions (if yes/no)

Example:

```
if (a > b)
```

Condition gives Boolean result: True or False

But values of `a` and `b` stored in binary.

Super Easy Memory Trick

Boolean = Decision

Can I go? Yes/No

Binary = Counting

How many? 0,1,10,11...

Final Summary

Expression	Boolean	Binary
1 + 1	1	10
Meaning of +	OR	Addition
Used in	Logic gates	Numbers

One Line Answer

Boolean asks Is it true?

Binary asks What number is it?